

Sree Chandana Madabhushi

PORTFOLIO

SELECTED WORKS

This portfolio collects projects in building physics, energy and climate-based design, life cycle assessment and computational design.

It is organised in three parts:

01 Building Performance & Sustainable Design

Simulation, optimisation and life cycle assessment, from a single room to a city district

02 Computational Design & Machine Learning

Mathematical form finding with origami and data-driven prediction of material behaviour

03 AI-Assisted Design & Digital Prototyping

Generative AI for circular construction and interactive web and app prototypes

Projects and page numbers on the right are clickable.

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01

BUILDING PERFORMANCE & SUSTAINABLE DESIGN

Simulation, optimisation and life cycle assessment, from a single room to a city district

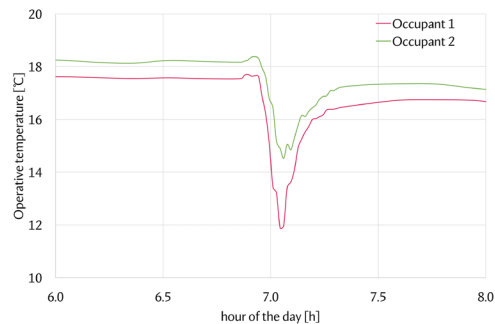
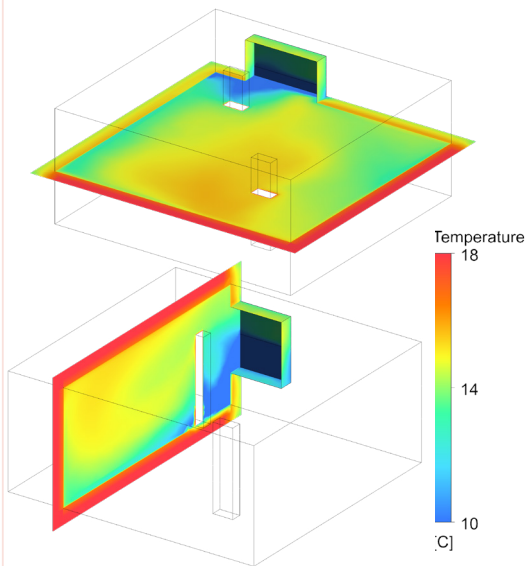
PROJECTS

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THERMAL HETEROGENEITY

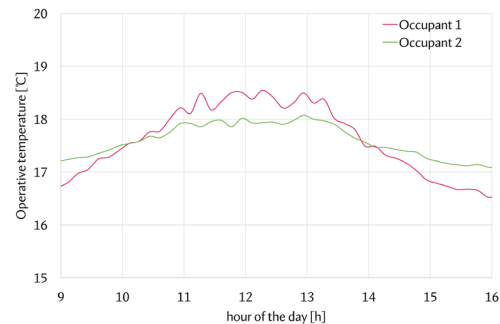
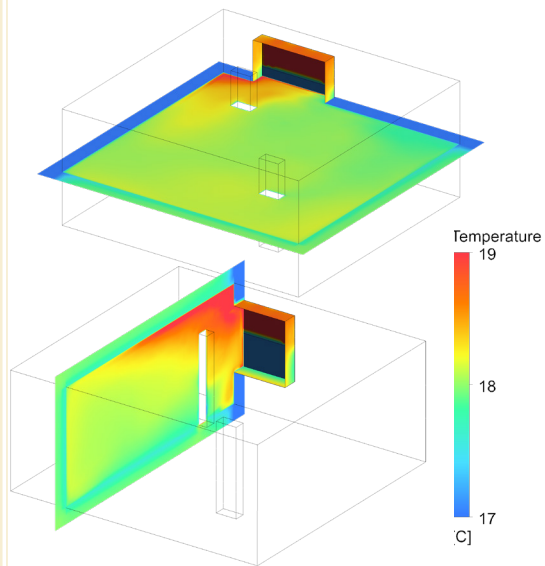
Ventilation

lets cold air in through window and causes thermal heterogeneity in the room until fully mixed.

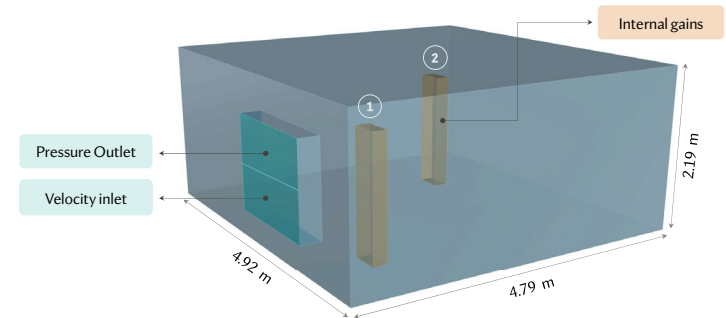
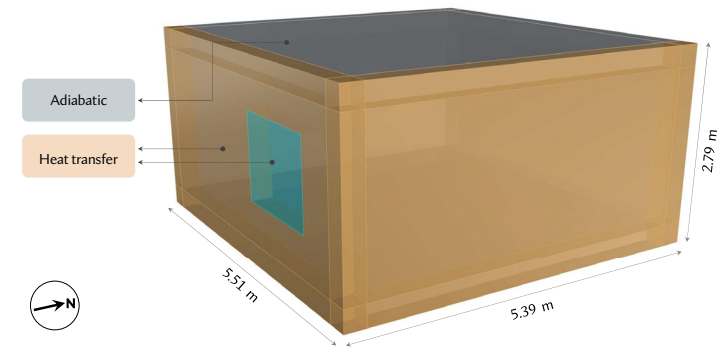
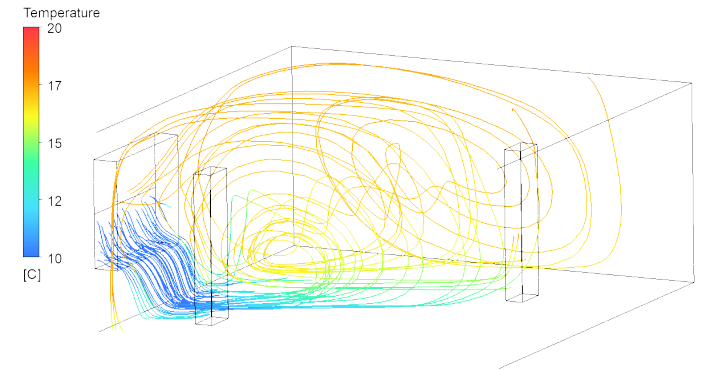


Solar radiation

causes heating up of window surface and thermal heterogeneity due to radiative heat near window.



using CFD



ENVELOPE BEHAVIOUR

METRICS

Operative Temperature

Temperature Gradient

Predicted Mean Vote

Wall Temperature

PHYSICS

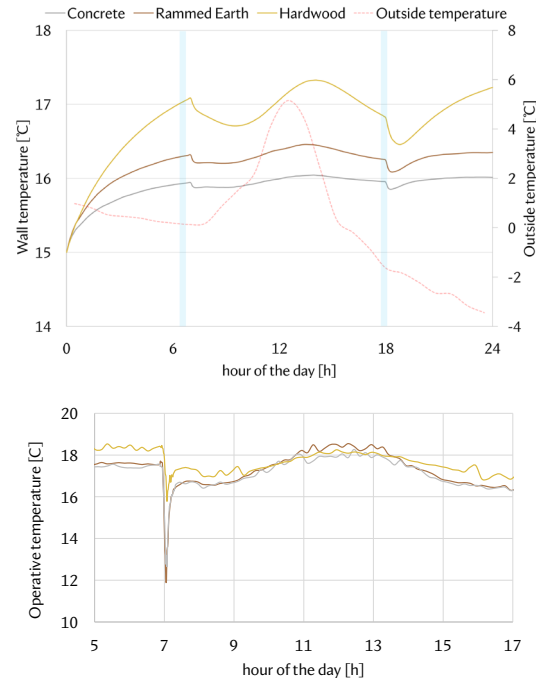
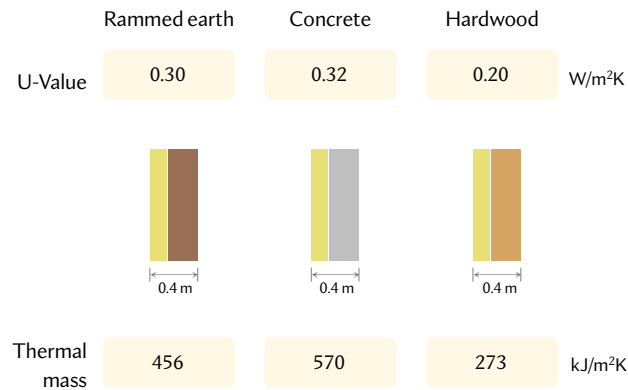
Heat transfer

Conduction through solid domain and convection through fluid domain are modelled. Solar radiation is approximated with a heatflux through window.

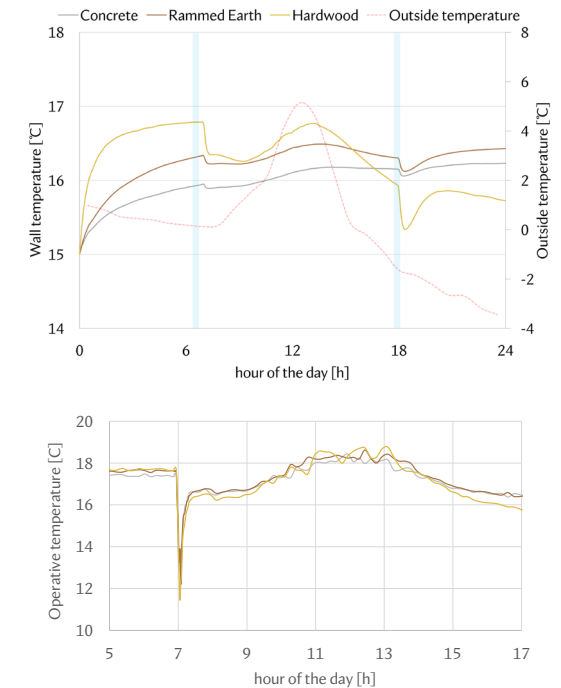
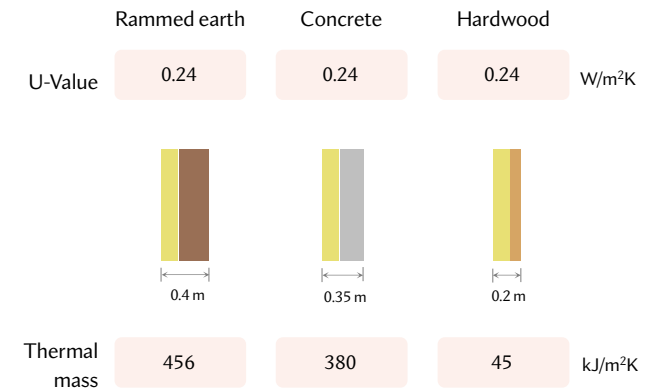
Fluid flow

Ventilation through window is modelled with velocity inlet in lower half and a pressure outlet in the upper half.

Equal thickness study

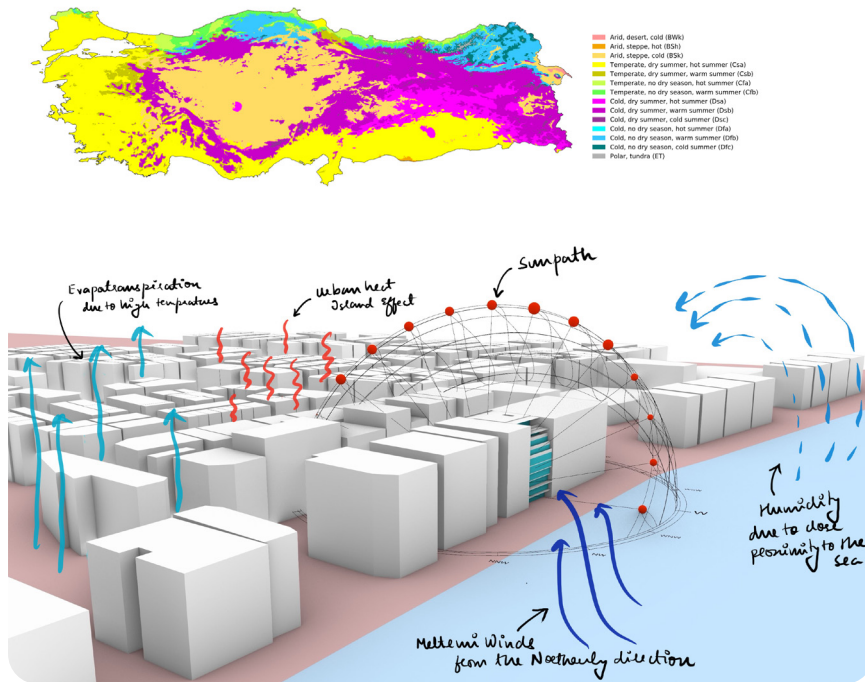


Equal U-Value study



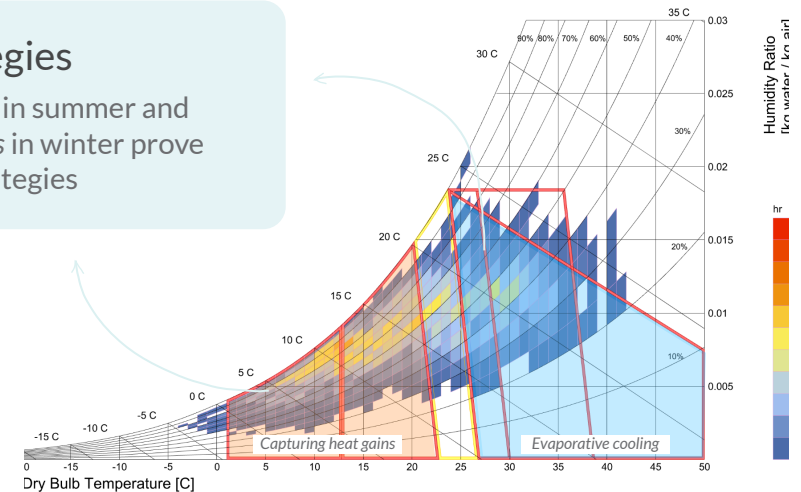
CLIMATE & ENERGY BASED DESIGN

An office building in Izmir, Turkey



Passive strategies

Evaporative cooling in summer and capturing heat gains in winter prove to be effective strategies



Summer discomfort

Summer discomfort is mainly due to higher indoor temperatures due to trapped internal heat gains

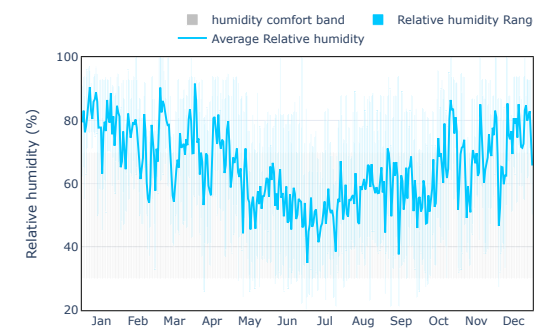
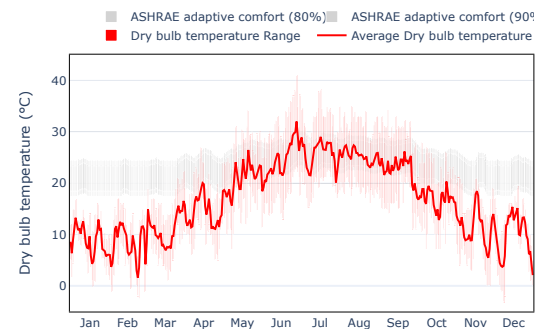
&

Winter discomfort

High relative humidity further lowers perceived temperature causing discomfort in winter

Mediterranean climate

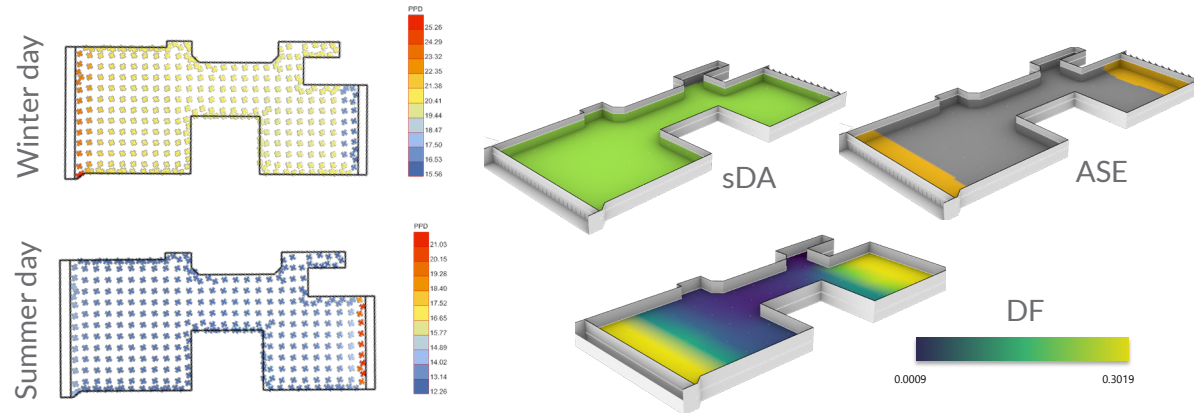
Dry warm summers and wet mild winters due to the shift of Azores high in the Atlantic. Proximity to the sea moderates summer temperatures.



OPTIMIZATION

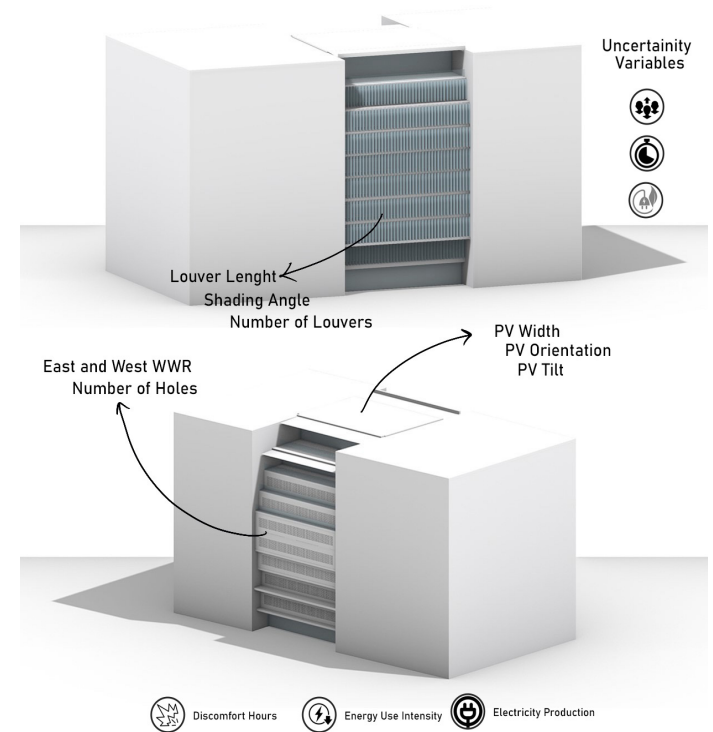
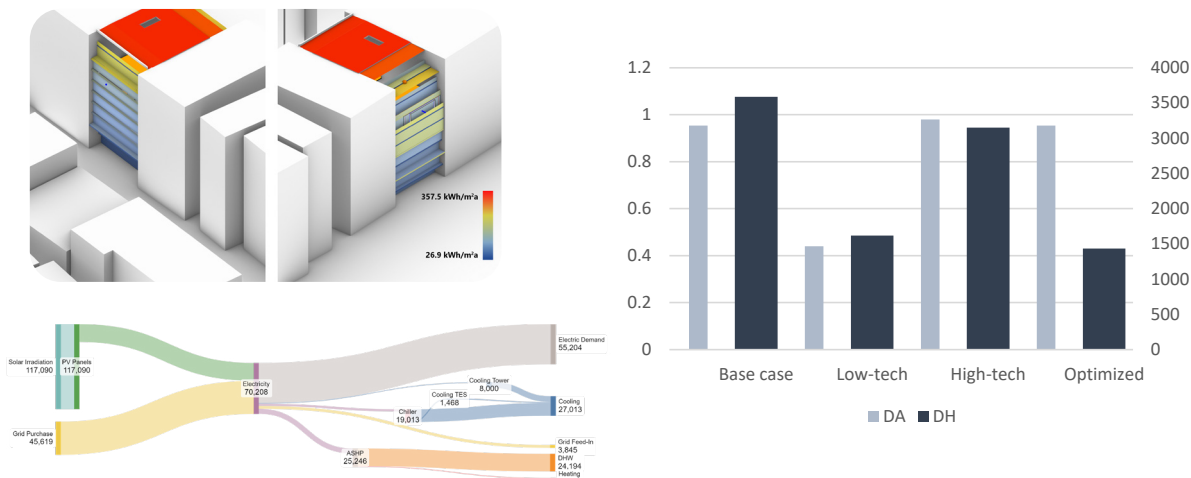
Minimize discomfort

in summer and winter using passive strategies



Maximize electricity generation

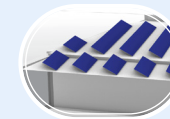
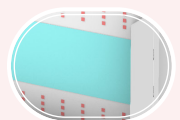
Optimized PV panel geometry and area for beneficial renewable energy generation



Vertical louvers

On east and west facades for shading

Wall screens
Mashrabiya-like screens for ventilation



PV Panels

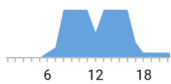
On the roof for renewable energy

URBAN ENERGY MODELING AND DESIGN

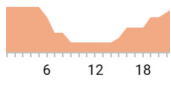
Techno-cultural city design in Shanghai, China

Mix-use

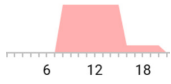
OFFICE



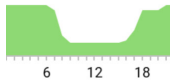
HOTEL



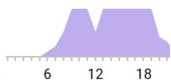
SCHOOL



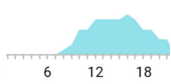
MULTI_RES



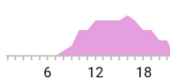
HUB



MUSEUM



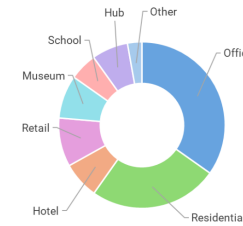
RETAIL



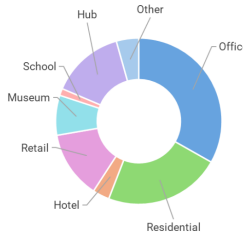
The city is designed to have high floor area ratio in 10 min circles from public transport. The image here shows the urban scape with solar radiation incident on the buildings.



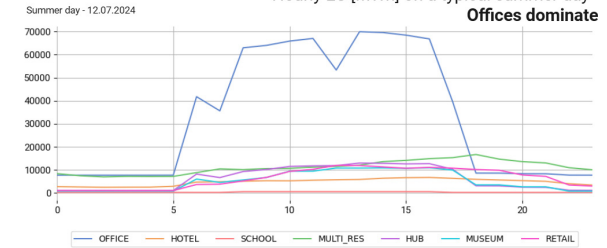
Energy use distribution reflects the use-mix



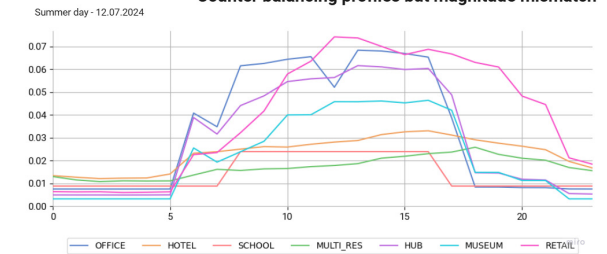
Energy use intensity distribution reveals high intensity use-types like hubs and retail



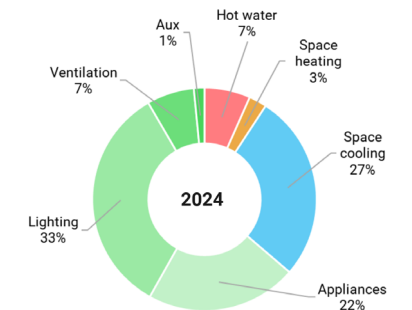
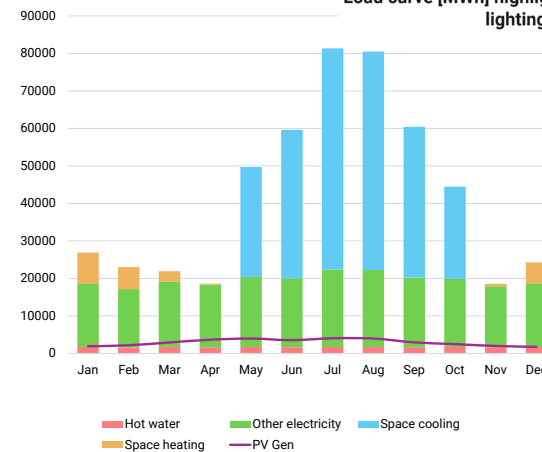
Hourly EU [MWh] on a typical summer day - Offices dominate



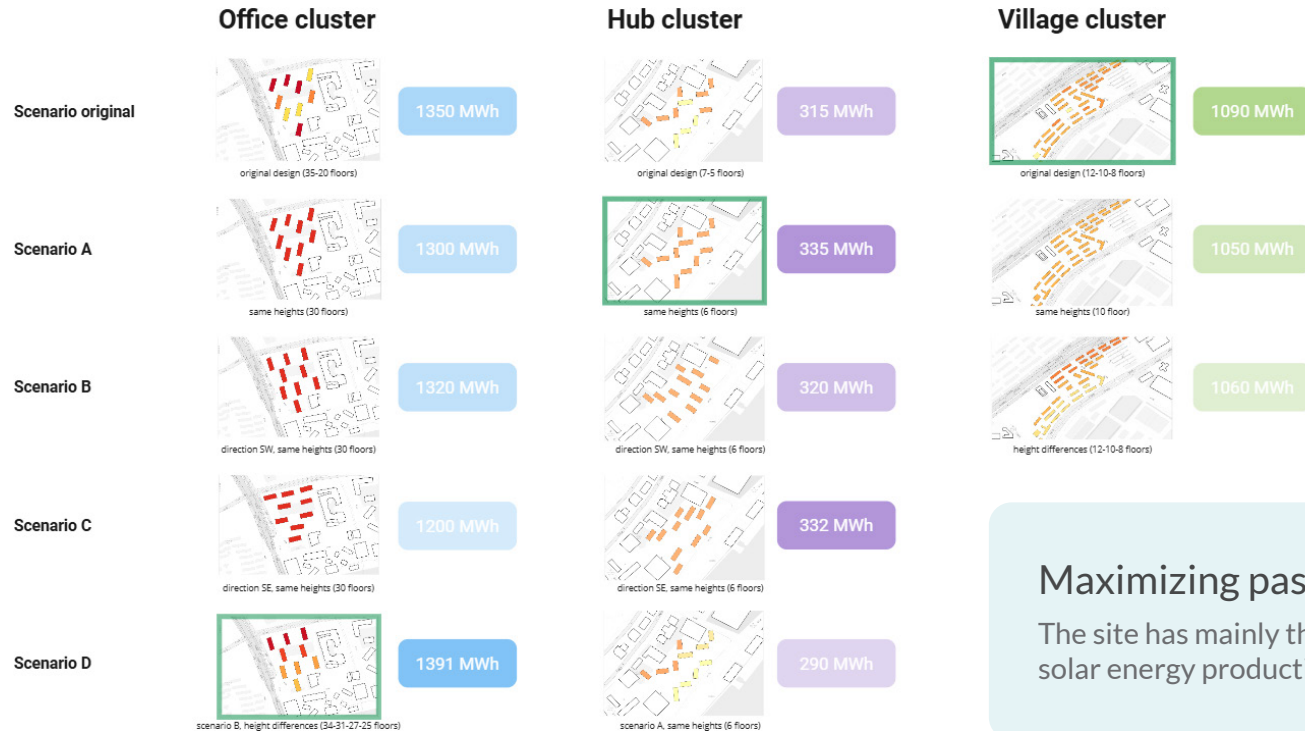
Hourly EUI [kWh/m2] on a typical summer day - Counter balancing profiles but magnitude mismatch



Load curve [MWh] highlights high cooling loads and high share of demand for lighting and appliances reflects the high share of office use.

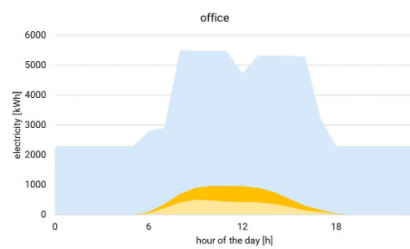


URBAN DESIGN OPTIMIZATION

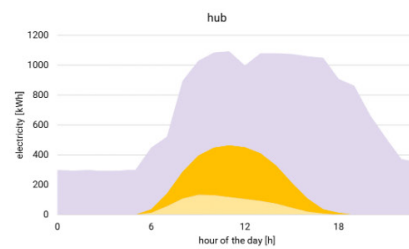


Maximizing passive energy potential

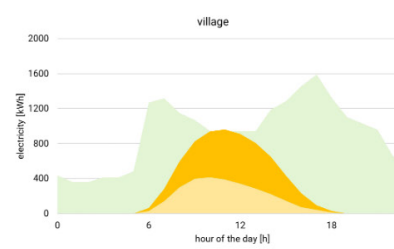
The site has mainly these three clusters, which are optimized to maximize solar energy production improving overall self sufficiency and consumption.



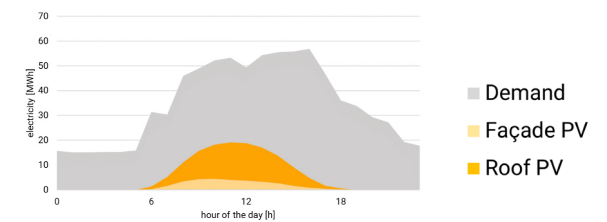
SS 9% SC 100%



SS 20% SC 100%



SS 28% SC 92%



SS 16% SC 100%

■ Demand
■ Façade PV
■ Roof PV

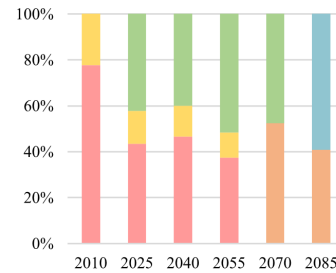
URBAN SCALE LCA AND FUTURE SCENARIOS

Embodied vs Operational CO₂

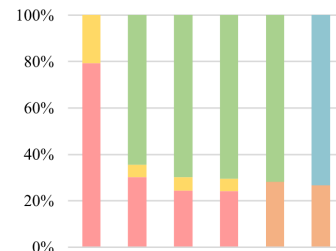
Embodied emissions remain insignificant until sufficient decarbonization of grid in Shanghai. Current focus needs to be on reducing the enormous operational emissions.

CdTe panels are eco-friendly

Low irradiation threshold indicates that installing CdTe panels is preferred over grid usage in terms of emissions.



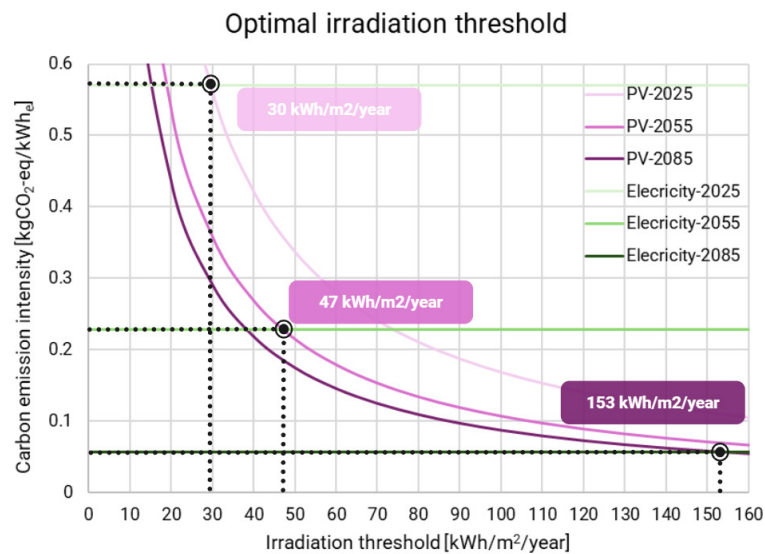
Embodied CO₂



Operational CO₂



The plot shows timeline of overall emissions considering potential interventions in buildings, decarbonization of grid etc..

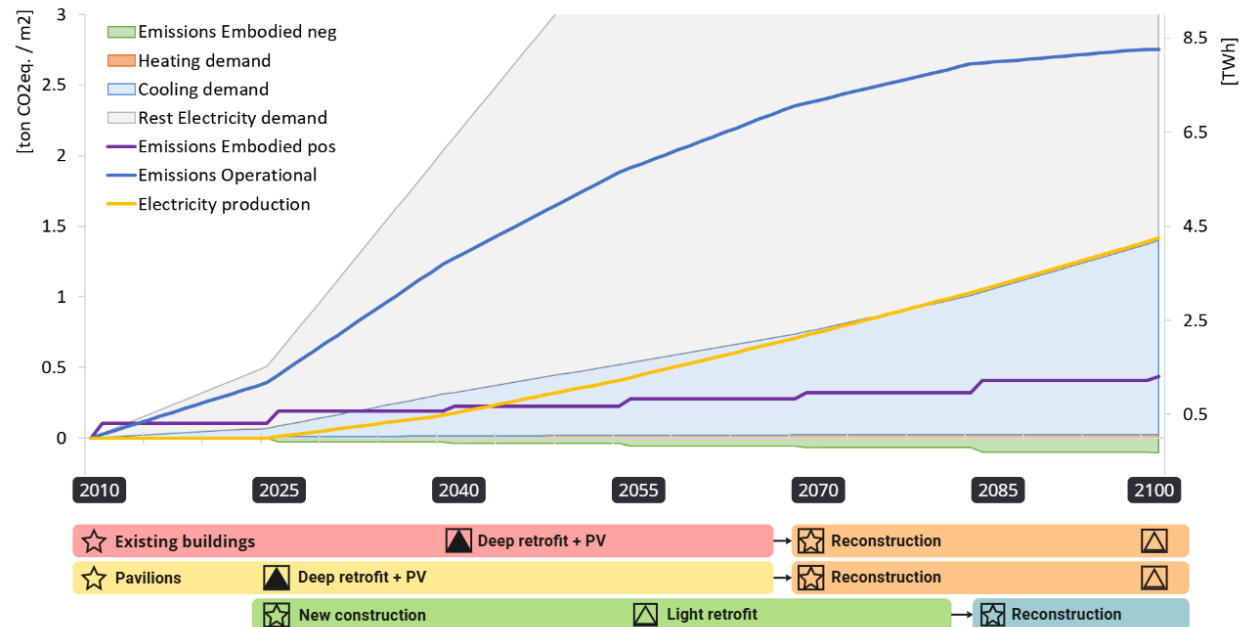


15% efficiency
75% performance ratio
25 year life span

20% efficiency
75% performance ratio
30 year life span

24% efficiency
75% performance ratio
30 year life span

PV embodied emission : 47.6 kgCO₂eq.



Existing buildings

Deep retrofit + PV

Reconstruction

Pavilions

Deep retrofit + PV

Reconstruction

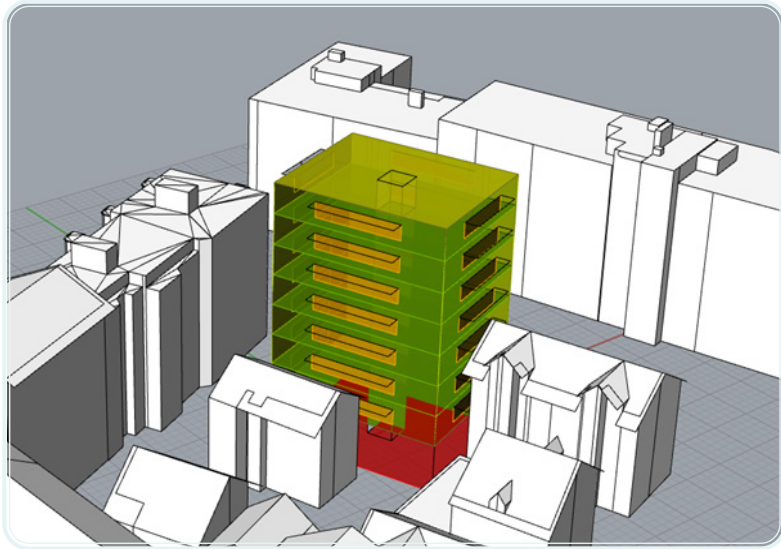
New construction

Light retrofit

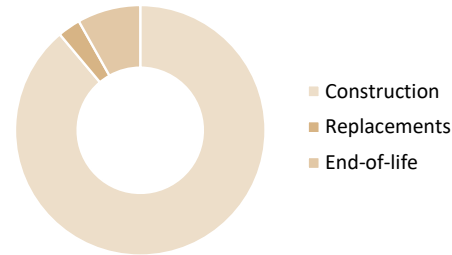
Reconstruction

DESIGN-INTEGRATED LCA

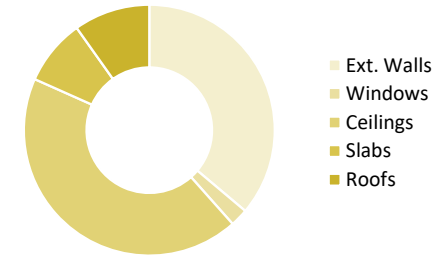
A parametric design case study



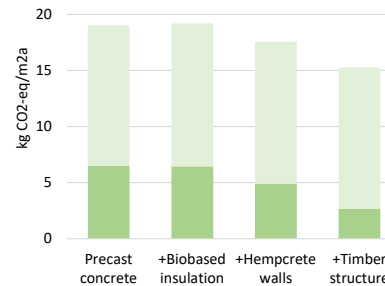
Embodied Emissions - Life Cycle Stage



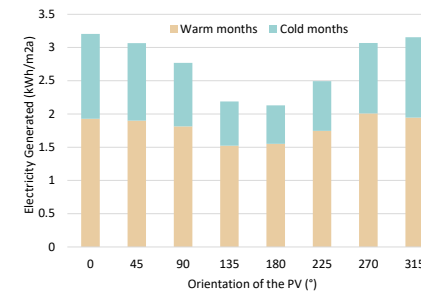
Embodied Emissions - Building Elements



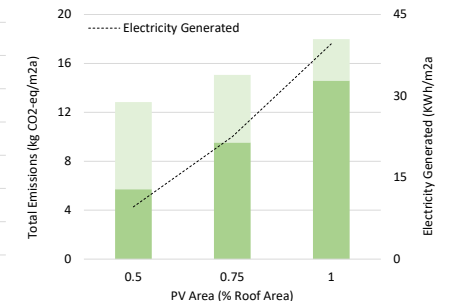
Material choice



PV panel geometry



Operational Emissions
Embodied emissions



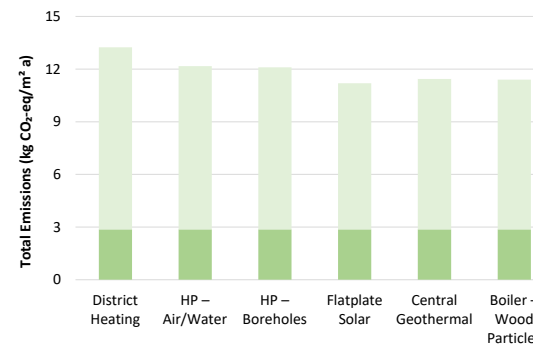
CO₂ emissions as an efficiency metric

Design is parametrized and optimized for minimizing emissions over its life cycle using Bombyx and Hive which uses KBOB database.

Sensitive parameters

Choice of these design parameters influence emissions in long-run (life cycle). However, the analysis is limited to only envelope and omits structural requirements.

Type of heating system

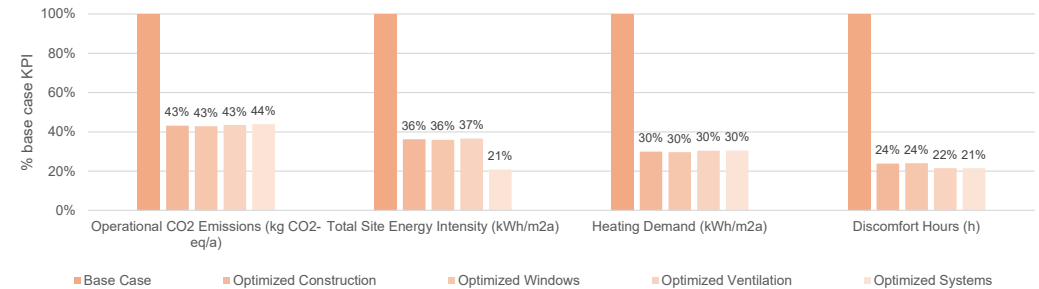
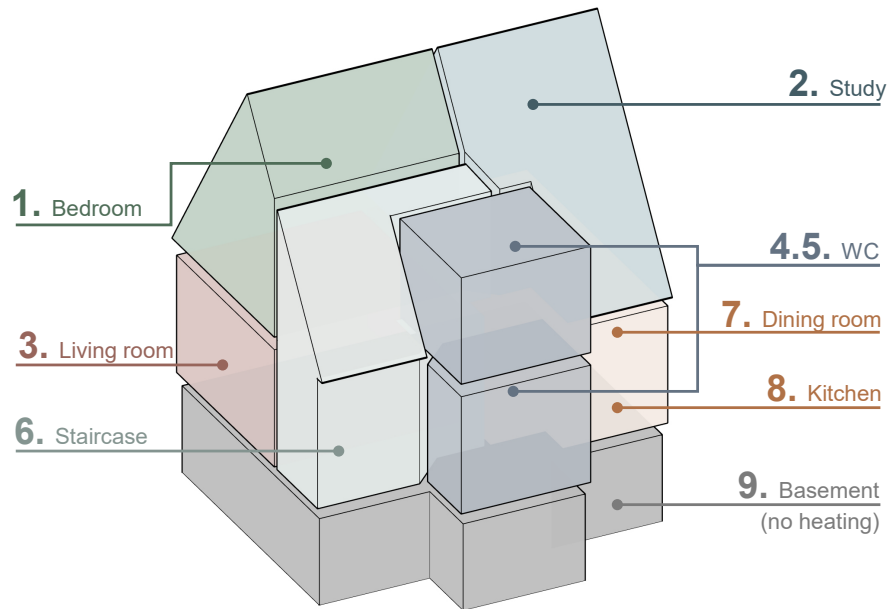


Window to wall ratio



RENOVATION STRATEGIES

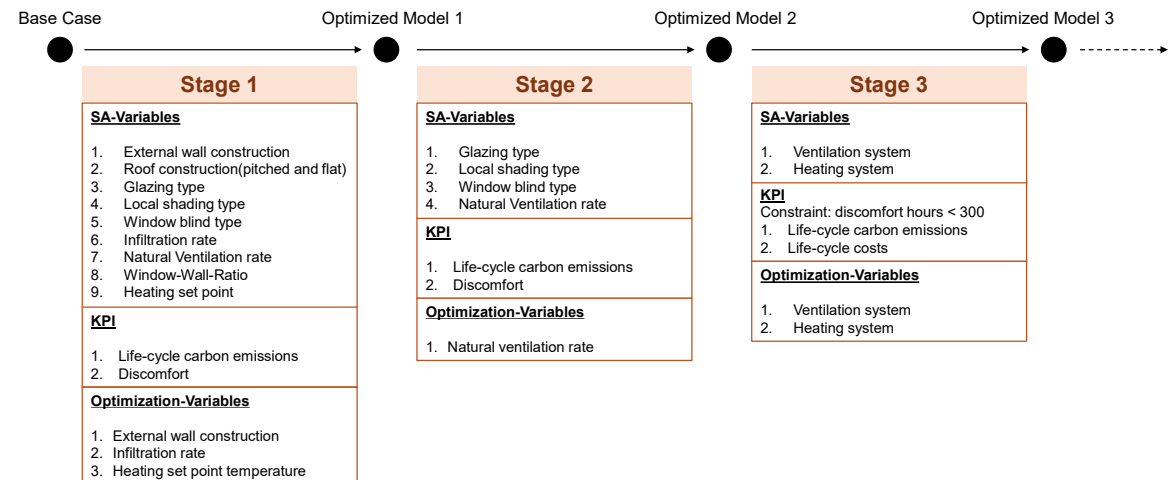
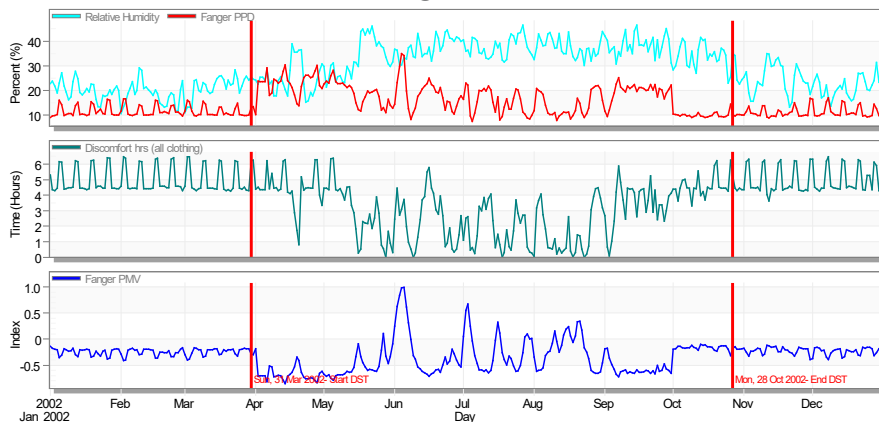
Case study of an old house in Olten, Germany



Sequence of building performance improvement

EnergyPlus simulation on base case showed poor performance, which was improved with a series of sensitivity analyses and optimization stages. The objective is mainly to reduce discomfort hours while minimizing life cycle carbon emissions and life cycle costs.

Discomfort results from DesignBuilder



02

COMPUTATIONAL DESIGN & MACHINE LEARNING

Mathematical form finding with origami and data-driven prediction of material behaviour

PROJECTS

Topographical Design using Origami

Design inputs · Engineering behind it (IASS 2020/21 paper)

13 – 14

Prediction of Glass Fracture Behaviour using ML

Variational autoencoder with neural cellular automata

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ORIGAMI

TOPOGRAPHICAL DESIGN

using Origami

Design Inputs

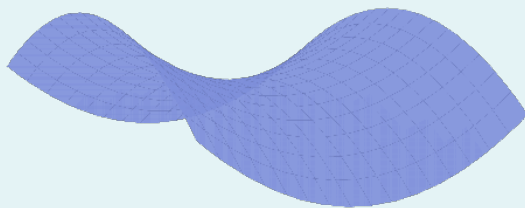


Origami Pattern

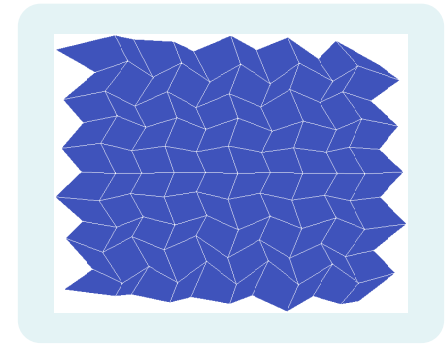
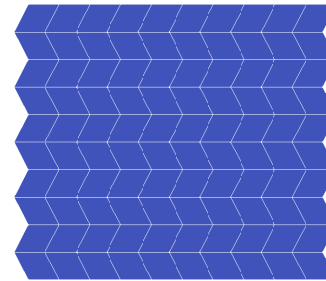
The standard Miura - Ori pattern, made of parallelograms, forms two parallel surfaces when folded along its crease pattern.

A 3D surface with arbitrary Gaussian Curvature, which the origami solution conforms to.

Target Surface



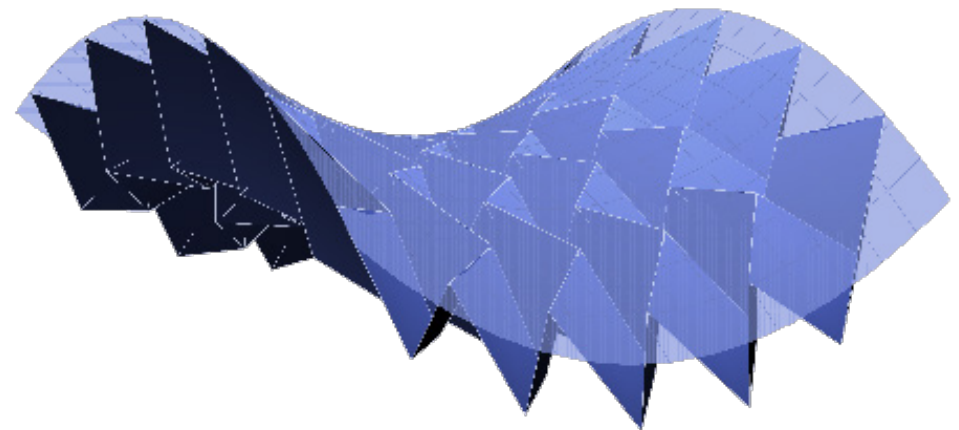
Folds into



Crease pattern

Starting with the standard Miura crease pattern, the algorithm aims to impose the design constraints and obtain a crease pattern that folds into the solution origami structure.

Folds into

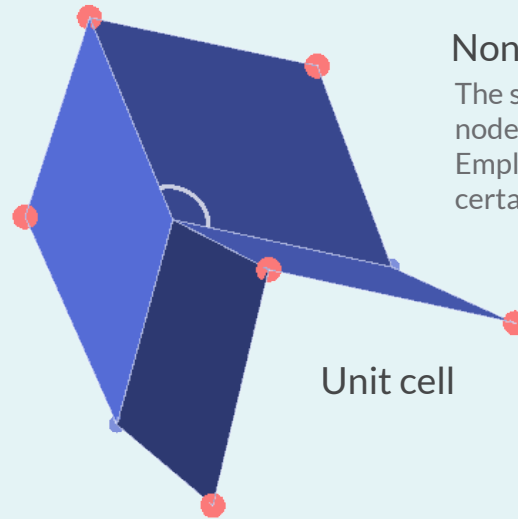
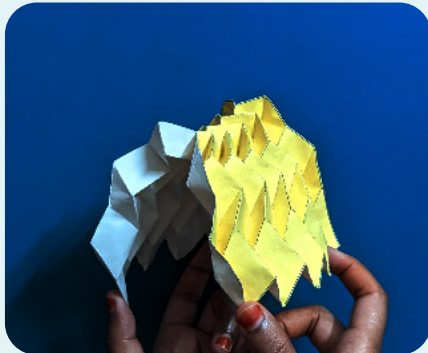


Engineering behind it!

RIGID

Functional constraints

- Developable from flat sheet.
- All faces remain rigidly planar.
- Structure is rigidly foldable.



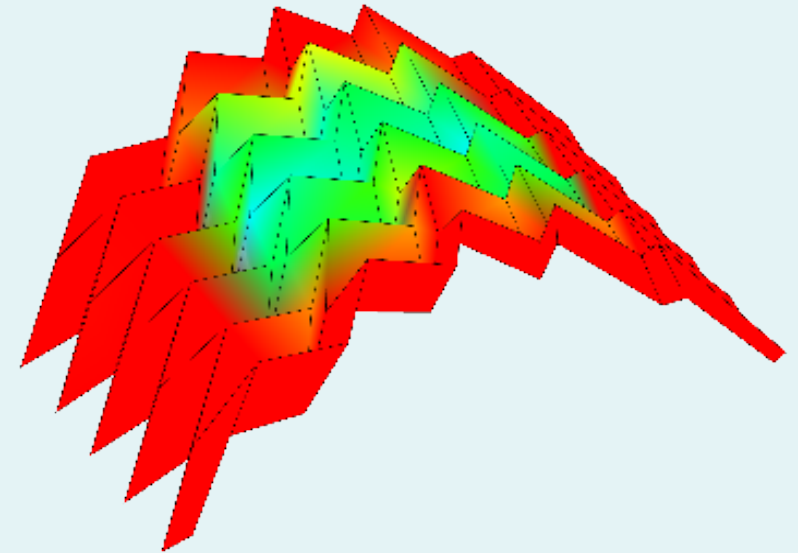
Unit cell

Non-linear system

The set of variables include co-ordinates of the pattern nodes, angles of faces and folds and size of the panels. Employed a least-squares based non-linear solver with a certain tolerance limit on deviations.

Strain visualization

Studied the rigid foldability of solutions by simulating the static strains developed during the folding motion and observed the same with physical models as well.



Publication

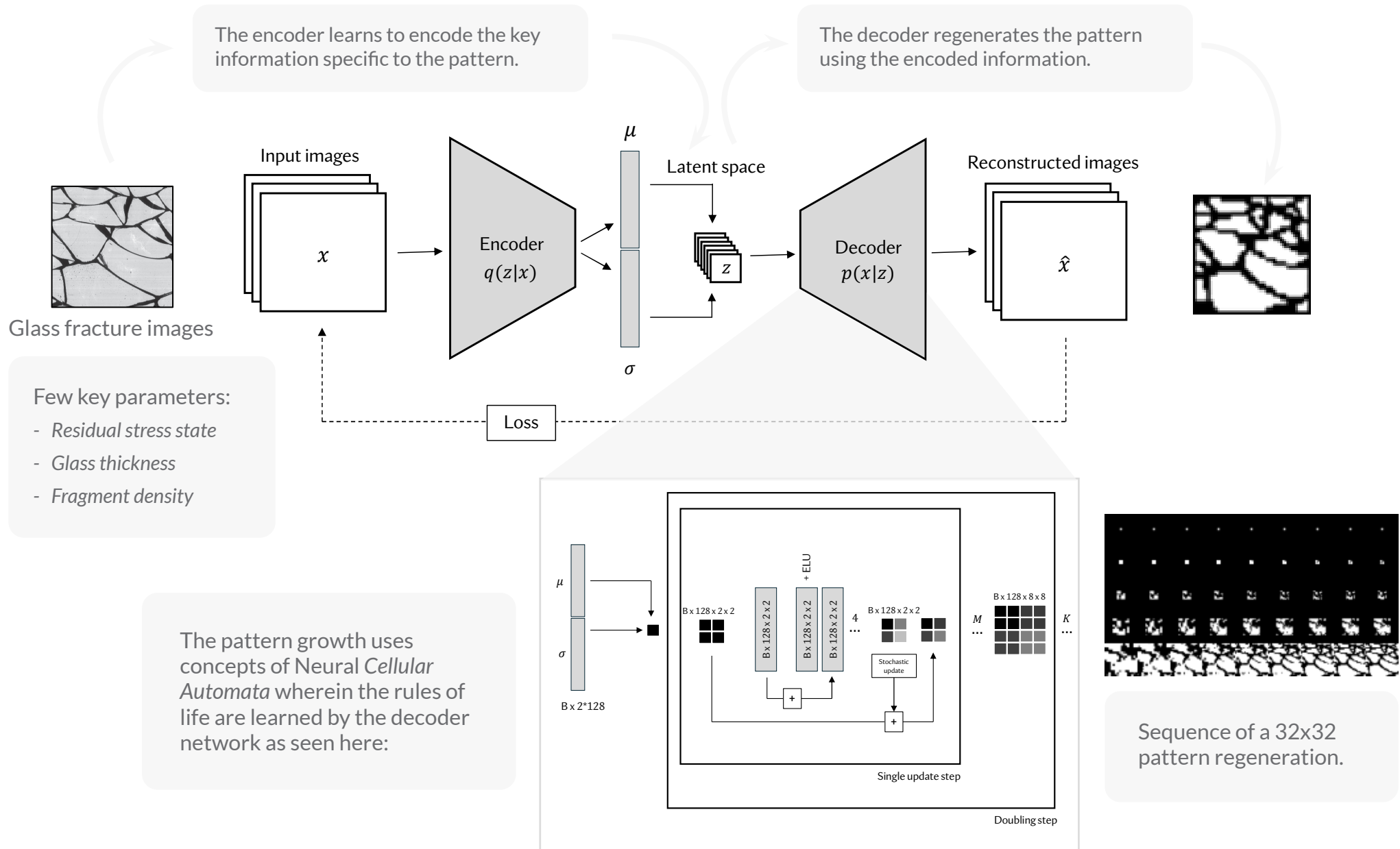
The work is published in the conference - [link](#).
Co-authored with Kishore S. Shenoy and
Dr. Phanisri Pradeep Pratapa.



**IASS 2020/21
SURREY 7**

International Conference
on Spatial Structures 2021/1

PREDICTION OF GLASS FRACTURE BEHAVIOUR USING ML



03

AI-ASSISTED DESIGN & DIGITAL PROTOTYPING

Generative AI for circular construction and interactive web and app prototypes

PROJECTS

AI in Design for Circular Construction

Interactive website prototype

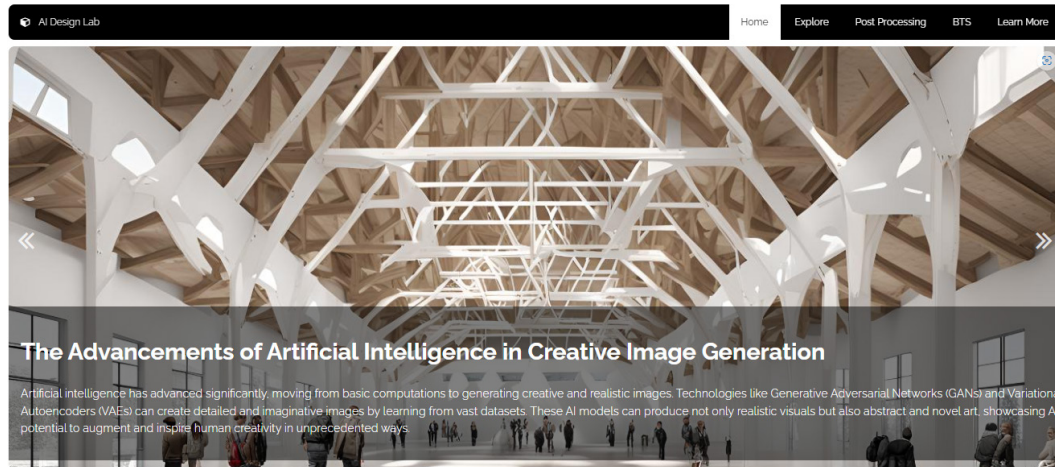
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UI/UX Design and Prototyping

Telemedicine app for a healthcare startup

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AI IN DESIGN FOR CIRCULAR CONSTRUCTION - WEBSITE



Explore



Midjourney

Contextual Design

An photograph from a contemporary with focus on the local . The structure must clearly show its elements and how they can be disassembled and repurposed.

Modularity in design

Create an interior space of a modular which could be assembled and disassembled easily. The concept of modularity within construction should be incorporated into the design. As the main building material should be used. The singular elements need to be distinguishable from one and each other.



Midjourney

We explored several concepts of circularity in design like modularity, etc., using AI image generators like DALL-E, Midjourney, and Runway ML. This website demonstrates the potential and limitations of these tools, by allowing users to tweak the prompt and providing a walk through of the process. Find the working prototype here - [link](#).

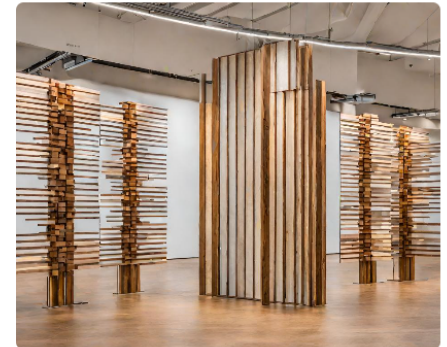
Post Processing



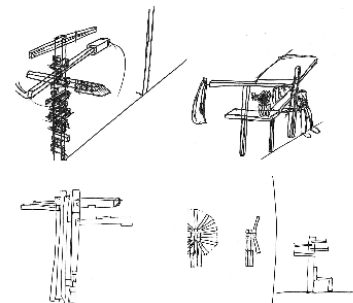
Stage 1: Generating Images using AI

Coat Hanger

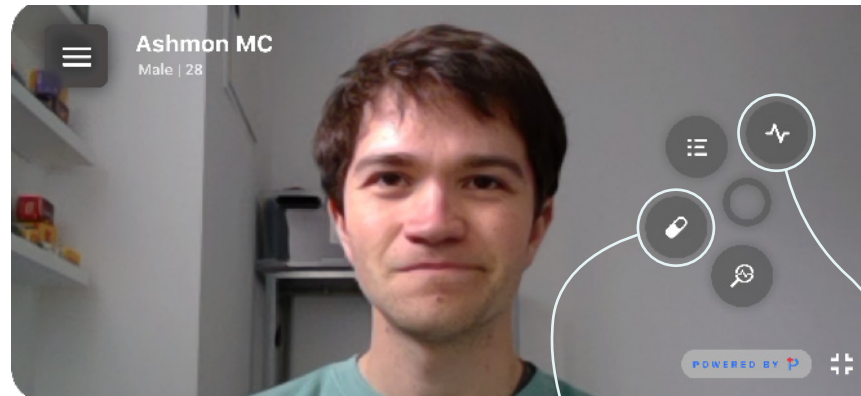
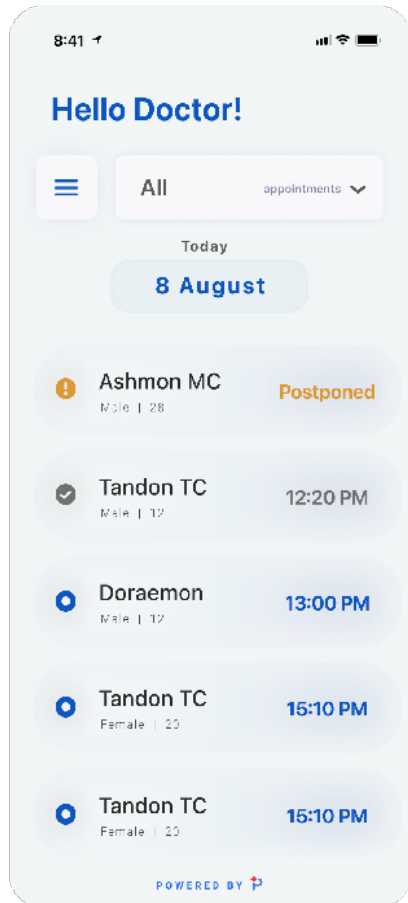
a modular futuristic exhibition wall made with reused polycarbonate planks and wooden beams



Stage 2: Sketching and prototyping



UI/UX DESIGN AND PROTOTYPING - WORKFLOW



Heatmap analysis shows accessibility and visibility of a particular frame.

Navigation icons



Final proceeding button is always at the bottom of the frame, to create a sense of ending and have a glimpse at the verge of clicking it.

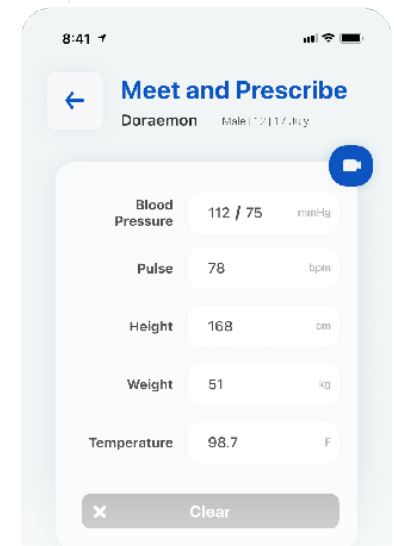
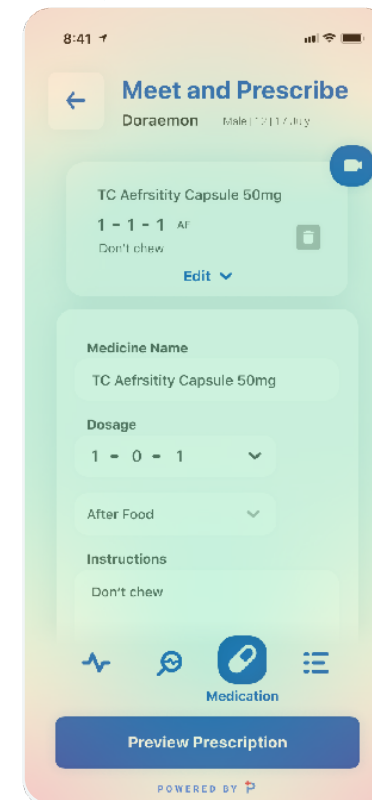
USER

The product is designed for a healthcare startup, that allows seamless doctor - patient consultations. Find the working prototype here - [link](#).

Most accessible wheel position. Draggable too!

+ Color scheme to indicate *medical* and go with the company logo.

x Grey for a sense of subtle warning.



Minimized video button for simultaneous interaction.

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